SESSION 5

Your Experiment Might Fail. Design It Anyway.

Design Thinking for MBAs

Saturday, April 4, 2026 | Instructor: Patrick Ray

Today's Arc

Phase	What Happens
Feedback	Gallery walk your Problem Definitions. Read, respond, sharpen.
Ideate	Generate genuinely different solutions. Frame them as testable hypotheses.
Stress Test	Audit your assumptions. Another team tries to break them.
Experiment	Design a test you can run in 5 days with zero dollars.

🧭 **Thursday night (Session 6):** You present what you learned, not what you planned. A killed idea is a successful experiment.

Is Your Problem Statement Narrow Enough?

A good problem statement passes all five:

1. **Can you name the specific people affected?**
   Not "employees" but "new nurses in their first rotation at a community hospital."
2. **Can you point to where it happens?**
   A specific place, process, or moment. Not a general condition.
3. **Could you observe it in a single visit?**
   If you'd need a month of data to even see the problem, it's too broad.
4. **Does your research support it?**
   Can you trace the statement back to something a stakeholder said or you directly observed?
5. **Could a team of five test a response in 5 days?**
   If the problem requires policy change or organizational restructuring to address, narrow further.

TOO BROAD

"New employees struggle to onboard effectively."

GETTING CLOSER

"New nurses don't know who to ask for help during their first rotation."

NARROW ENOUGH

"New nurses at community hospitals rely on informal hallway conversations to learn which unwritten protocols vary by unit, because no formal handoff exists between orientation and floor work."

Problem Definition Gallery Walk

Setup

1. Print and **post your Problem Definition** on the wall at your team's station
2. Include your revised causal loop map
3. One team member stays as **docent** (answers questions, does not present)
4. Everyone else circulates

~30 minutes total

Feedback Framework

Leave written notes at each station:

I Like...

What's sharp? Where is the evidence strongest?

I Wish...

What's still vague? What needs tightening?

I Wonder...

What question does this raise? What's missing?

Solutions Tic-Tac-Toe

Draw a 3x3 grid on your large paper.

Fill every cell with a solution concept. Each must be **mutually incompatible** with the others. No variations on a theme.

All 9 cells must be filled when time is up.

20 minutes

	Individual	System	Policy
Low Risk	*Peer mentor program for new hires*	*Shared onboarding checklist across teams*	*Require 30-day check-in with every new employee*
Med Risk	?	?	?
High Risk	?	?	?

Example: Scenario A (The First 90 Days)

Frame Your Hypothesis

Pick your **1-2 strongest concepts** from the grid. Reframe each as a testable hypothesis:

> "We believe **[solution]** will **[outcome]** for **[stakeholder]** because **[insight from our research]**."

The "because" clause is the test. It must trace to something a stakeholder actually said or did during your research. If it doesn't point to evidence, you're guessing.

Write your hypotheses on large paper and **post them on the wall.** Then walk the room and read every team's hypothesis silently. No presenting.

~30 minutes (framing + posting + reading)

Assumption Audit

You just read each other's hypotheses. Now find the weakest link in your own.

Team Activity

1. What makes you **nervous** about your own hypothesis?
2. List **5 critical assumptions** it depends on
3. Rank by lethality: **"If this is wrong, our concept fails"**
4. For your top 2: what **specific, observable evidence** would disprove this?

~15 minutes

Confidence without evidence is just optimism. If two people can't agree on whether your kill criterion has been met, it's not specific enough.

Adversarial Stress Test

Find weaknesses, not strengths.

1. **Half your team stays** at your table as defenders. Other half walks to another team's table as attackers.
2. **Attackers:** find the assumption that, if wrong, wastes their five days.
3. **12 minutes.** Then swap roles at a different table.

~25 minutes (12 min each round)

Questions worth asking:

- "You're assuming the stakeholder wants this. Do you know that, or do you believe that?"
- "What evidence do you have beyond your own intuition?"
- "What if your problem definition is wrong?"
- "Who benefits if this works, and who doesn't?"

Problem Check

Did anything you just heard change how you understand **the problem**, not just the solution?

Silent write in your notebook (2 minutes, individual). Then share with your team. Revise your hypothesis if needed.

~10 minutes

Three Tools, One Company

Before Thursday, you need to run an experiment. Here's what that means, and what it doesn't.

MVP

Built to Run

A product you ship and keep running. It's rough, but it works as a service. You grow it into a business.

SURVIVAL TACTIC

Built to Not Die

Pure scrappiness. Whatever it takes to keep the lights on and prove you won't quit.

PROTOTYPE

Built to Learn

Tests one assumption with one real person. Might be disposable. Not built to last. Built to find out if you're right.

You're building a prototype. Not an MVP. Not a business. A test.

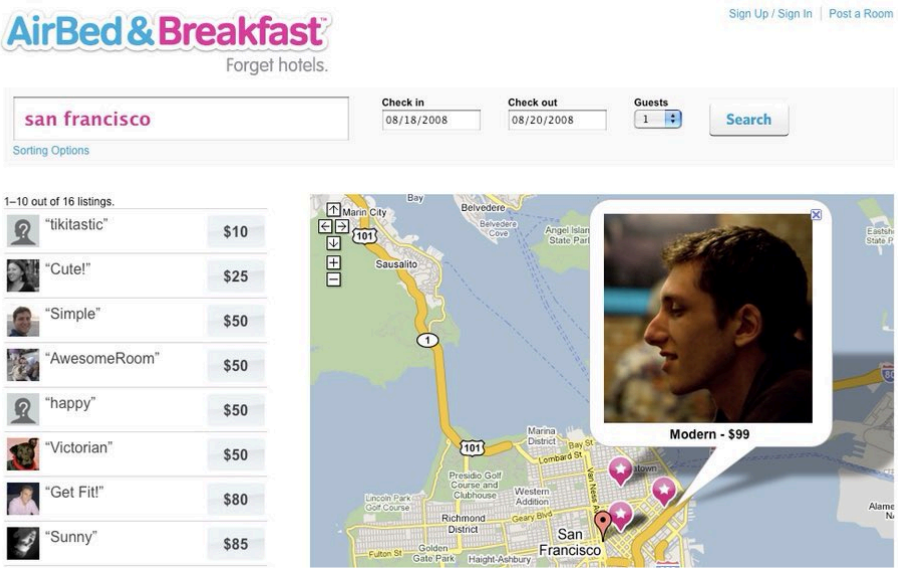
The MVP: Air Mattresses, 2007

MVP

Chesky and Gebbia can't make rent. Design conference in SF, hotels sold out. They buy three air mattresses, build a simple website: AirBed and Breakfast. $80/night. Breakfast included.
Three strangers show up and pay.

They kept running it. Kept hosting.

It became the business.



The Survival Tactic: Cereal Boxes, 2008

SURVIVAL

One year later. Airbnb is nearly dead. No funding, no traction, massive debt. The 2008 election is happening.

They design custom cereal: Obama O's and Cap'n McCains. 500 boxes each. $20 to make, $40 to sell. They sell out. $30,000. Enough to keep the lights on.

> **Paul Graham, Y Combinator:**
> "If you can convince people to pay $40 for a $4 box of cereal, you can convince people to stay in each other's homes. You are cockroaches. You will not die."

The Problem: Would You Book This?

The Prototype: Photography, 2009

PROTOTYPE

Airbnb has users, but nobody's booking. Hypothesis: the photos are terrible and that's killing trust.
They rent a camera, fly to New York, knock on hosts' doors. One week. One apartment at a time.

2-3x more bookings.
Revenue doubled that month.

Then they threw away the prototype and built the real program.

> "A web startup would say, 'Let's send emails, teach users professional photography, and test them.' We said, 'Screw that.'"
>
> "When we fixed the product in New York, it solved our problems in Paris, London, Vancouver, and Miami."
>
> Chesky & Gebbia, Fast Company

Design Your Experiment

Team Activity: Answer four questions. Write them on your large paper.

1. **What's the riskiest assumption you're testing?**
2. **What's the smallest thing you can do to test it?**
   Not the whole solution. Just the test.
3. **What does "pass" look like? What does "fail" look like?**
   Specific enough that two people can agree on whether it happened.
4. **Who specifically will you test with, and when?**
   Name a real person. Pick a day.

~45 minutes

Plus: task assignments. Each team member's name, their specific action, their stakeholder, the day they'll do it.

NEW CONSTRAINT

Your first test must cost zero dollars.

Experiment Plan Review

Poster Session

1. Your experiment plan stays at your table.
2. **Half your team stays as defenders.** Other half circulates.
3. At each table, ask only two questions.
4. **Swap at 12 minutes** so everyone both presents and visits.

~24 minutes

Question 1

"Is this testable in 5 days?"

Question 2

"What decision does this test let you make?"

You have 5 days.
Go test it.

Session 6: Thursday, April 9 | 6:30 PM | Online

Present what you learned and who taught you.
Not a polished pitch. The story of what happened.

A killed idea is a successful experiment.

Notebooks due Sunday, April 12, 11:59 PM